Retail Transaction Data Integrity & Violation Detection Policy

1. Purpose

This policy defines the standards, validation rules, and violation detection mechanisms for transactional invoice data collected within the Retail Sales Management System. The purpose of this policy is to:

- Ensure data accuracy and consistency.
- Detect fraudulent or erroneous transactions.
- Maintain financial and reporting integrity.
- Prevent misuse of customer and pricing information.
- Enable automated compliance validation during data ingestion and processing.

This policy applies to all records containing:

- Invoice identifiers
- Product stock codes
- Product descriptions
- Quantities
- Invoice timestamps
- Unit prices
- Customer identifiers
- Country information

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2. Scope

This policy applies to:

- CSV uploads
- Database insert operations
- API ingestion endpoints
- Data warehouse synchronization
- Analytical processing systems

All transaction records must comply with the validation rules described below before being stored or processed.

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3. Data Field Validation Rules

3.1 Invoice Number (InvoiceNo)

Rule 1: Format Validation

- InvoiceNo must be:
  - Numeric (e.g., 563613) OR
  - Alphanumeric starting with "C" (e.g., C563600) indicating cancellation.

Rule 2: Cancellation Rule

- If InvoiceNo starts with "C":
  - Quantity must be negative.
  - It must correspond to a previously issued positive invoice.

Violations:

- Negative quantity with non-cancellation InvoiceNo.
- Cancellation invoice with positive quantity.
- Duplicate InvoiceNo with conflicting transaction types.

3.2 StockCode

Rule 1: Non-null Constraint

- StockCode cannot be empty.

Rule 2: Consistency Rule

- Each StockCode must consistently map to the same Description.
  - Example: StockCode 22845 must always represent the same product.

Violations:

- Same StockCode mapped to multiple descriptions.

- Non-alphanumeric or malformed StockCode.
- Missing StockCode.

3.3 Description

Rule 1: Mandatory Field

- Description must not be empty.

Rule 2: Consistency

- Description must match the registered product catalog for the given StockCode.

Violations:

- Description mismatch for same StockCode.
- Suspicious characters or injection patterns.
- Overly long or truncated descriptions.

3.4 Quantity

Rule 1: Positive Quantity

- Quantity must be positive for normal invoices.
- Quantity must be negative only for cancellation invoices (InvoiceNo starting with "C").

Rule 2: Threshold Rule

- Maximum single transaction quantity limit:
    - Example threshold: 1000 units.
    - Transactions above threshold require manual review.

Rule 3: Zero Quantity

- Quantity cannot be zero.

Violations:

- Zero quantity.
- Positive quantity in cancellation invoice.

- Extremely large quantity (possible fraud or data error).
- Negative quantity without cancellation InvoiceNo.

Example from your dataset:

C563600 ... Quantity: -18 → Valid (cancellation)
563613 ... Quantity: 100 → Valid (bulk purchase)

3.5 UnitPrice

Rule 1: Positive Value

- UnitPrice must be strictly greater than 0.

Rule 2: Price Range Check

- UnitPrice must fall within acceptable product pricing range.
- Sudden deviations > 300% of historical average must trigger alert.

Rule 3: High-Value Monitoring

- UnitPrice above predefined threshold (e.g., 500) requires review.

Violations:

- Zero or negative price.
- Extreme price anomaly.
- Price mismatch for same StockCode across invoices.

3.6 InvoiceDate

Rule 1: Valid Date Format

- Must follow valid datetime structure.

Rule 2: Logical Date Constraint

- InvoiceDate cannot be:
    - In the future.
    - Before system inception date.

Rule 3: Sequence Validation

- Cancellation InvoiceDate must be later than original invoice.

Violations:

- Future timestamp.
- Invalid date format.
- Backdated manipulation attempts.

3.7 CustomerID

Rule 1: Mandatory Field

- CustomerID must not be null.

Rule 2: Numeric Format

- CustomerID must be numeric.

Rule 3: Risk Monitoring

- Excessive bulk purchases by same CustomerID in short timeframe must trigger alert.

Violations:

- Missing CustomerID.
- Non-numeric ID.
- Suspicious high-frequency transactions.

3.8 Country

Rule 1: Approved Country List

Country must belong to predefined allowed list:

- United Kingdom
- EIRE
- Israel
- Australia

- Channel Islands
- etc.

Rule 2: Country-Customer Consistency

- CustomerID should not frequently switch countries without valid reason.

Violations:

- Unknown country.
- Suspicious country switching pattern.

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4. Violation Categories

4.1 Minor Violations

- Formatting errors
- Missing optional metadata
- Minor description mismatch

Action: Logged, auto-correct if possible.

4.2 Moderate Violations

- Price anomalies
- High quantity threshold breaches
- Duplicate invoice detection

Action: Flagged for review before approval.

4.3 Severe Violations

- Fraud pattern detection
- Cancellation abuse
- Negative price manipulation
- CustomerID spoofing

Action:

- Transaction blocked
- Alert compliance team
- Temporary customer suspension

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5. Automated Violation Detection Logic

When new CSV is uploaded:

1. Schema validation.
2. Field-level validation.
3. Cross-record validation:
    - Check duplicate InvoiceNo.
    - Validate StockCode consistency.
    - Compare price history.
4. Fraud scoring engine:
    - High quantity + high price.
    - Frequent cancellations.
    - Rapid country switching.

If violation score > threshold:

- Record marked as "Under Review"
- Not included in reporting totals.

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6. Cancellation Abuse Detection

Because your dataset shows invoices like:

C563600 → negative quantities

The system must ensure:

- Each cancellation references a valid original invoice.

- Total cancellation quantity does not exceed original purchase quantity.
- Repeated cancellation behavior per customer is monitored.

Violation example:

- Customer buys 10 units.
- Cancels 20 units.
  → Fraud attempt.

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7. Data Storage Policy

- All validated records stored in primary transaction collection.
- Violations stored in separate violations_log collection with:
  - InvoiceNo
  - ViolationType
  - Severity
  - Timestamp
  - ReviewStatus

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8. Audit & Monitoring

- Weekly anomaly reports generated.
- Monthly fraud trend analysis.
- Annual pricing integrity audit.

Logs retained for minimum 5 years.

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9. Compliance Enforcement

Failure to comply with this policy may result in:

- Transaction rejection
- Account suspension

- Manual audit
- Legal reporting (if fraud confirmed)

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10. Conclusion

This Data Integrity & Violation Detection Policy ensures:

- Clean financial reporting
- Fraud detection
- Customer accountability
- Accurate analytics
- Secure transaction processing

By implementing these rules at API level before database insertion, the system maintains high-quality transactional data and reduces operational risk.